WILLOW

Video-based batting analysis for cricket coaches and aspiring players. Project guide, October 2026.

One line: a coach films a student's batting, the app reads body positions from the video, scores nine parts of the technique 0-100, writes plain-English feedback, and tracks each student over time. Invited athletes get their own Strava-style progress view.

1. What is built

Coach side (finished for v1)

- Sign up and log in; add, edit and archive students (batting hand, under-18 flag).
- Upload batting clips (resumable, private, up to 50 MB and 60 s), tagged with student, date and shot type.
- Automatic analysis in the browser: pose tracking, nine scored metrics, overall score, written feedback, body-outline overlay on the video.
- Coach can override any metric score, mark clips reviewed, and add notes (private, or shared with the athlete).
- Student dashboard: trend chart, latest vs previous vs best per skill, personal bests, warnings.
- Roster overview: who needs attention (declining, unreviewed, inactive) and top improvers. Demo data button.

Athlete side (invite-only, built and tested except analysis)

- Coach invites a student by email; the invite works only for that email and expires after 14 days.
- Athlete home, upload page, and result page (coach-adjusted scores win; only shared notes are visible).
- Strava-style progress page: week-vs-last card, trend chart, 8-week streak calendar, personal bests, skill trends.
- Not yet tested: athlete self-analysis (needs the server secret key saved). Not built: feed cards, reminders, guardian consent.

2. What we did, in order

Stage	What happened
Plan	Chose Next.js + Supabase + in-browser pose tracking + AI only for wording; defined the nine batting metrics and the data model so the athlete side could be added without rewrites.
Setup and login	Coach sign-up and login on Supabase, with database rules so each coach only ever sees their own students.
Roster and uploads	Student roster; private, resumable video upload with progress bar and size/length limits.
Analysis	Built pose tracking and scoring. Fixed along the way: 309 s to 25 s per clip, a model mode that found no one, jumping between people (now follows one), camera shake (now corrected), and body parts cut off (now left unscored with a warning).
Results and tracking	Score breakdown, coach overrides, reviewed flag, notes, student dashboard with trends, roster overview, demo data, and the body-outline overlay on the video.
Privacy check	Manual two-coach test passed: the second coach saw nothing of the first coach's students or videos.

Stage	What happened
Design	Four visual directions prototyped (Aurora, Zinc, Clay, Kinetic); you chose Kinetic and the name Willow; applied to the whole app.
Athlete side	Roles and rules with a 43-check privacy test (all passed); coach invites by email; athlete home, upload and results; Strava-style progress page (designed first, then built, maths checked).
Release prep	Production build verified; developer-only pages locked in production; deployment plan written (DEPLOY.md).

3. How the scoring works

Pose tracking (MediaPipe) finds body keypoints in the browser; plain code turns them into numbers; an AI model only phrases the feedback from those numbers. The AI never sees the video and never guesses metrics. Distances are measured in torso lengths so camera distance and body size do not matter.

Metric	Weight	What is measured
Head stillness	15%	Head bob and sideways drift from set-up to contact
Stance balance	12%	Hips centred over the feet before the shot
Head over front knee	12%	Head position relative to the front knee at contact
Swing path	12%	How direct the hands travel from backlift to contact (bat itself is not tracked)
Footwork	12%	Front-foot travel, compared with the shot type
Timing	12%	Bat contact relative to front-foot landing
Weight transfer	10%	Hip shift towards the bowler, compared with the shot type
Backlift height	8%	Hand height at the top of the backlift
Follow-through	7%	How far the hands rise after contact

Important: every threshold is a first-pass heuristic, not validated against real coach scores. Treat scores as estimates. To calibrate, have a cricket coach score about 30 clips per shot type, compare with the app, then tune the ranges (sports/cricket/shots.ts and batting.ts) and weights.

Footage needs: phone propped steady, side-on, whole body in frame with space above the head, one batter, about 10 seconds. The app warns or declines to score when the body is cut off or the camera moves a lot.

4. Privacy and safety

- Videos live in a private storage bucket; they are only viewable through links that expire after an hour. No public URLs.
- Every table has row-level security: a coach sees only their own students; an athlete sees only their own record, clips and results.
- Athletes cannot write or change scores, mark clips reviewed, or move clips to another student (database trigger + rules).
- A 43-check privacy test (supabase/tests/athlete-privacy-test.sql) passed, and a manual two-coach test passed.
- Known limit: body keypoints are computed in the athlete's browser, so a technical athlete could fake them. Planned fix: re-analyse on the server.
- Before real students: guardian consent for under-18s, privacy notice, separate production database, rate limit on analysis, security review.

5. Tech and how to run

Next.js 16 (TypeScript, Tailwind) + Supabase (login, Postgres, private storage) + MediaPipe in the browser + OpenAI for feedback wording. Look: "Kinetic" (off-white, ink black, electric blue, hot orange; Anton, Archivo, Space Mono).

Step	Command / action
Install	npm install
Database	Supabase SQL Editor: run supabase/schema.sql, then supabase/athlete-side.sql
Keys	Copy .env.example to .env.local; fill Supabase URL + public key, OpenAI key, and (for athlete analysis) the Supabase secret key
Run	npm run dev, then open http://localhost:3000
Try it	Sign up, Load demo data, open a student, upload a clip, Analyse

6. Going live

Target: the existing Vercel project at hackathon-byob3.vercel.app (currently the old soccer app). Full checklist in DEPLOY.md. Summary:

- Tag the old soccer app's last commit; Vercel's instant rollback also restores it.
- Enter five environment variables in Vercel (secret key and OpenAI key are secrets).
- Add the live address in Supabase Authentication URL settings; turn email confirmation on and add custom email sending before real users.
- Merge cricket-coach into main and push; test the live link end to end.

7. Open items

Item	Owner	Status
Save Supabase secret key (athlete analysis)	Founder	Waiting (terminal tab 'Save secret key')
Test scoring on a good steady side-on clip	Founder	Needs footage
Cricket coach calibrates scores (about 30 clips)	Founder + coach	Not started
Guardian consent + privacy notice (under-18s)	Build	Next
Check name 'Willow' (app stores, domain, trademark)	Founder	Not checked
Deploy to Vercel	Both	Plan ready (DEPLOY.md)